

CoRoT-2b

R-1527

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_150709 · created 2026-08-01T17:47:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457212.8294 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.063
Transit depth (Rp/Rs)^2	2.7e-05 +/- 0.0064 [%]
Orbital Inclination (inc)	89.62 +/- 2.06
Airmass coefficient 1 (a1)	40.2 +/- 23.36
Airmass coefficient 2 (a2)	-3.0 +/- 0.83
Scatter in the residuals of the lightcurve fit is	65.9 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	3.0
Transit Duration (day)	0.0844 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.063 vs 0.1667 (deviation 2.63σ)
Duration fit vs published	0.0844 d vs 0.095 d (deviation 2.3σ)
Mid-time O-C	-4.2 min
Depth SNR	0.0
Residual scatter	65.9 %

Light curve

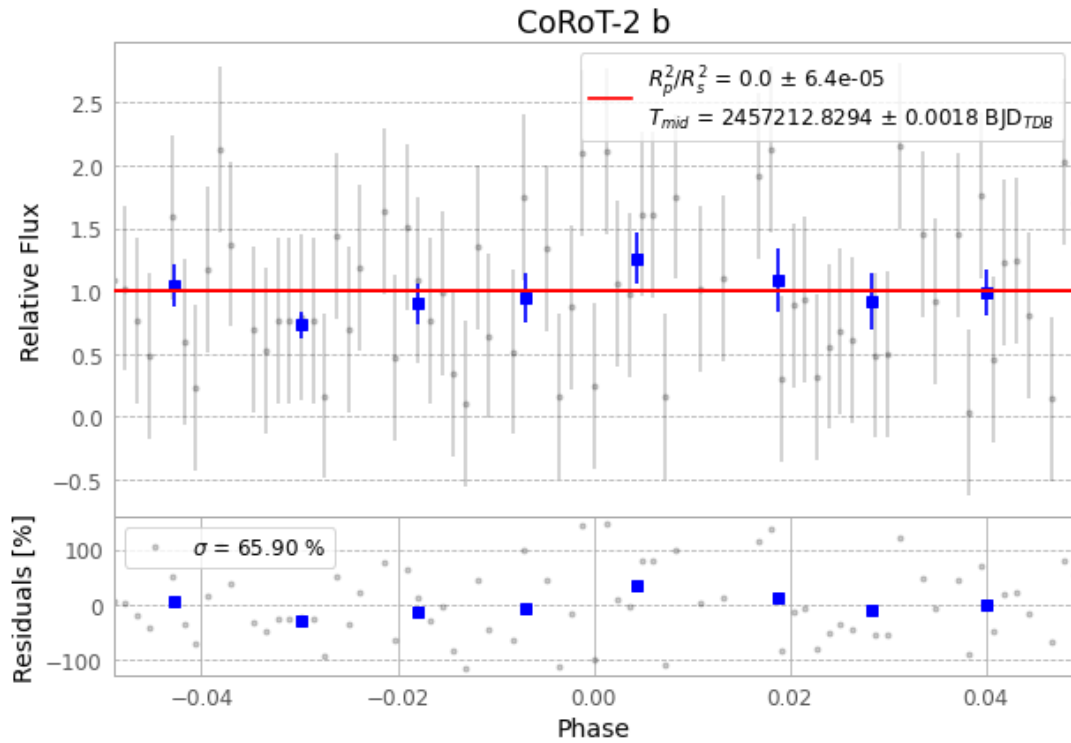


Figure 1 — FinalLightCurve_CoRoT-2 b_2015-07-08.png (SHA-256 4010803834cd527d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [413, 194], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c8acf310a271cfcf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), CoRoT-2150709054212.FITS → CoRoT-2150709095712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-150709.log (d77d709204c1...), FinalLightCurve_CoRoT-2 b_2015-07-08.png (4010803834cd...), FinalLightCurve_CoRoT-2 b_2015-07-08.csv (d572c231800e...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 17:47Z.